

[Aim: 100/100 in Maths]

आरंभ

TRIGONOMETRY

Lecture #6

ATK²
Reciprocal of

$$\sin \theta \longleftrightarrow \frac{1}{\operatorname{cosec} \theta}$$

$$\cos \theta \longleftrightarrow \frac{1}{\sec \theta}$$

$$\tan \theta \longleftrightarrow \frac{1}{\cot \theta}$$

$$\left[\begin{array}{l} \tan \theta = \frac{\sin \theta}{\cos \theta} \\ \cot \theta = \frac{\cos \theta}{\sin \theta} \end{array} \right]$$

Square of formula

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\begin{cases} 1 - \cos^2 \theta = \sin^2 \theta \\ 1 - \sin^2 \theta = \cos^2 \theta \end{cases}$$

Similar brackets

#LP: Prove that $(\sin A + \cos A + 1)(\sin A - 1 + \cos A) \sec A \operatorname{cosec} A = 2$

LHS

$$\Rightarrow ((\overset{a}{\sin A} + \overset{b}{\cos A}) + (1)) ((\overset{a}{\sin A} + \overset{b}{\cos A}) - (1)) \sec A \operatorname{cosec} A$$

$$\Rightarrow \left[(\overset{a^2}{\sin A + \cos A})^2 - (\overset{b^2}{1})^2 \right] \sec A \operatorname{cosec} A$$

$$\Rightarrow [\sin^2 A + \cos^2 A + 2 \sin A \cos A - 1] \sec A \operatorname{cosec} A$$

$$\Rightarrow [\cancel{1} + 2 \sin A \cos A - \cancel{1}] \sec A \operatorname{cosec} A$$

$$\Rightarrow (2 \cancel{\sin A} \cancel{\cos A}) \frac{1}{\cancel{\cos A}} \times \frac{1}{\cancel{\sin A}}$$

$$\Rightarrow \textcircled{2} = \text{RHS} \quad \textcircled{\cancel{\text{HP}}}$$

$$\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix} \right) \sin^2 A + \cos^2 A = 1$$

जब जब अलग - अलग दिखे $\rightarrow$ सबको $\sin, \cos$ में आओ

#LP: Prove that $(\operatorname{cosec} A - \sin A)(\sec A - \cos A)(\tan A + \cot A) = 1$

→ T-ratios
 $\sin, \cos$ में
 एल ट्राइ
 convert
 करें

$\sin \theta$

$\cos \theta$

$$\tan \theta = \frac{s}{c}$$

$$\operatorname{cosec} \theta = \frac{1}{s}$$

$$\sec \theta = \frac{1}{c}$$

$$\cot \theta = \frac{c}{s}$$

$$\Rightarrow \left(\frac{1}{s} - s \right) \left(\frac{1}{c} - c \right) \left(\frac{s}{c} + \frac{c}{s} \right)$$

$$\Rightarrow \left(\frac{1-s^2}{s} \right) \left(\frac{1-c^2}{c} \right) \left(\frac{s^2+c^2}{cs} \right)$$

$$\Rightarrow \left(\frac{c^2}{s} \right) \left(\frac{s^2}{c} \right) \left(\frac{1}{cs} \right)$$

$$\Rightarrow 1 = \text{RHS} \quad \text{✓}$$

#LP: Prove that $\cot \theta - \tan \theta = \frac{2\cos^2 \theta - 1}{\sin \theta \cos \theta}$

RHS से आये $\sin, \cos$

LHS

$$\frac{c}{s} - \frac{s}{c}$$

$$\rightarrow \frac{c^2 - s^2}{sc}$$

$$\rightarrow \frac{c^2 - (1 - c^2)}{sc}$$

$$\Rightarrow \frac{c^2 - 1 + c^2}{sc} \Rightarrow \frac{2c^2 - 1}{sc}$$

= RHS

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

QED

#LP: $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$

(CBSE 2018)

LHS

$$\frac{\sin A (1 - 2\sin^2 A)}{\cos A (2\cos^2 A - 1)}$$

$$\Rightarrow \tan A \frac{(1 - 2\sin^2 A)}{(2\cos^2 A - 1)}$$

$$\Rightarrow \tan A \frac{(1 - 2(1 - \cos^2 A))}{(2\cos^2 A - 1)}$$

$$\Rightarrow \tan A \frac{(1 - 2 + 2\cos^2 A)}{(2\cos^2 A - 1)}$$

$$\Rightarrow \tan A \frac{(2\cos^2 A - 1)}{(2\cos^2 A - 1)}$$

$$\Rightarrow \tan A = \text{RHS} \quad \underline{\underline{1P}}$$

Q.P.:- Prove that $\frac{\cos \theta - 2\cos^3 \theta}{\sin \theta - 2\sin^3 \theta} + \cot \theta = 0$

LHS

$$\frac{\cos \theta - 2\cos^3 \theta}{\sin \theta - 2\sin^3 \theta} = -\cot \theta$$

$$\frac{\cos \theta}{\sin \theta} \frac{(1 - 2\cos^2 \theta)}{(1 - 2\sin^2 \theta)}$$

$$\Rightarrow \cot \theta \frac{(1 - 2(1 - \sin^2 \theta))}{(1 - 2\sin^2 \theta)}$$

$$\Rightarrow \cot \theta \frac{(1 - 2 + 2\sin^2 \theta)}{(1 - 2\sin^2 \theta)}$$

$$\Rightarrow \cot \theta \frac{(-1 + 2\sin^2 \theta)}{(1 - 2\sin^2 \theta)}$$

$$\Rightarrow -\cot \theta \frac{(1 - 2\sin^2 \theta)}{(1 - 2\sin^2 \theta)}$$

$$\Rightarrow -\cot \theta = \text{RHS} \quad \text{HP}$$

#LP: Prove that $\frac{\tan A}{1 + \sec A} - \frac{\tan A}{1 - \sec A} = 2 \operatorname{cosec} A$

HW

K3B

#QP: If $\sec \theta + \tan \theta = 4$, then $\sec \theta - \tan \theta$??

(A) 2

(B) 16

✓ (C) $\frac{1}{4}$

(D) $\frac{1}{2}$

we know,

$$\sec^2 \theta - \tan^2 \theta = 1$$

$a^2 - b^2$

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$\sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$

$$\sec \theta - \tan \theta = \frac{1}{4}$$

~~K3B~~

एक दूसरे के Reciprocal

$$\checkmark \left[\sec \theta + \tan \theta \longleftrightarrow \frac{1}{\sec \theta - \tan \theta} \right]$$

$$\checkmark \left[\operatorname{cosec} \theta + \cot \theta \longleftrightarrow \frac{1}{\operatorname{cosec} \theta - \cot \theta} \right]$$

$$\checkmark \left[\sin \theta + \cos \theta \longleftrightarrow \frac{1}{\sin \theta - \cos \theta} \right]$$

eg
if $\operatorname{cosec} \theta - \cot \theta = y$
then, $\operatorname{cosec} \theta + \cot \theta = \frac{1}{y}$

#LP: $\sec A (1 - \sin A) (\sec A + \tan A) = 1$

LHS $(\sec A - \sec A \cdot \sin A) (\sec A + \tan A)$

$\Rightarrow (\sec A - \frac{1}{\cos A} \sin A) (\sec A + \tan A)$

$\Rightarrow (\cancel{\sec A} - \cancel{\tan A}) \left(\frac{1}{\cancel{\sec A} - \cancel{\tan A}} \right)$

$\Rightarrow 1 = \text{RHS}$
HP

THANK YOU COODIESS !!

